Model-Driven-Engineering in Education

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***Abstract*— Students are taught in undergraduate lectures about dynamic systems, control theory and computer programming. However, in the engineering practice a more diverse skillset is required. In addition to the topics covered in lectures, Model Driven Engineering (MDE) should be introduced to the curriculum. With the help of MDE students can learn to develop controls for dynamic systems, furthermore they learn about signal processing and communication systems, using an efficient and effective method. As a case study, the controlled object that is used in this educational project is an Electronic Throttle Valve (ETV). The ETV is used as a laboratory project, because it can be controlled by a diverse range of controller designs, from simple to complex. During the design phase of a new controller, its functionality and reliability should be tested in all development phases. An appropriate process for testing and applying MIL, SIL, PIL and HIL simulation is the V-cycle.**

This paper describes the related education methods at Deggendorf Institute of Technology (DIT). As an introduction a short description of related educational activities at DIT in current study plans will be given. Thereafter, the modelling method applied to the ETV will be presented together with the control design by Input-Output-linearization. Comparison of the MIL, SIL, PIL and HIL simulation techniques for identifying potential errors are performed for each test-case. The test result of the hardware implementation on an FPGA board will be discussed in the last section.

Keywords—Mechatronics, Education, Throttle-Valve, Model-Based-Design, Position Control, Rapid Control Prototyping, X-in-the-Loop, FPGA.

I. Education Activities in Classes

A widely used approach in engineering design – especially in the automotive field – is the V-cycle shown in figure 1. The first branch of the V-cycle covers the model and control design, and for that the students need a theoretical background in control theory (which is part of engineering study plans of DIT). The right branch of V-cycle covers testing and verification of the control system. Student projects are also carried out according to the phases of the V-cycle, to transfer practical knowledge which can be applied in an industrial context.

A. Modelling and parameter identification

In theoretical classes, the basic elements of MDE [1,2] are mediated. The students are taught to model the physical system and describe it in time- and frequency-domain and state-space description, in order to perform the system identification and parameter estimation. Students can understand which physical variables should be measured and recorded for system identification. Furthermore the required knowledge for creating a simplified plant model is covered in the curriculum.

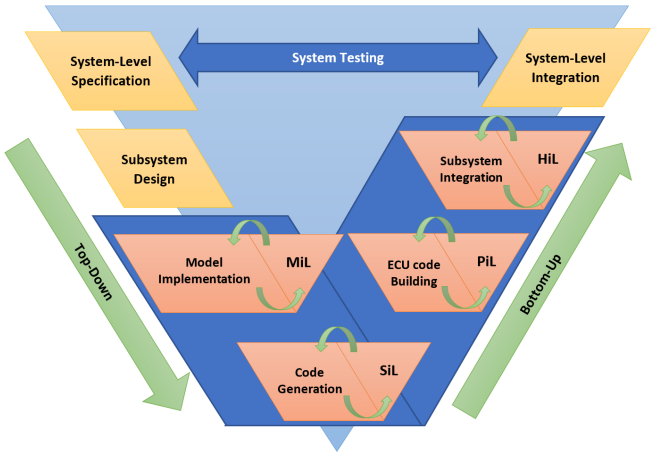

Fig. 1. Model based design according to the V-cycle.

B. Testing and Validation

After obtaining the required theoretical knowledge, the students are able to use a model for data analysis, visualization, testing and validation. They will understand the principle of code generation using the simulation model. Afterwards they recognize the reasons for the usage of the V-cycle instead of the traditional waterfall model. The stages of the modelling and control design are covered with following elements:

- Model in the Loop: MIL specifies that the model and controller are simulated on a computer. The simulation can then be used to diagnose errors in the control design and to adjust the model accordingly.
- Software in the Loop: SIL refers to a type of testing which is executed to verify the functionality of the generated production code. After successful design of the controllers, students can test their automatically generated code. SIL allows verifying the code coverage, and arithmetic errors (effects due to fixed-point variables and scaling).
- Rapid Control Prototyping: after satisfactory simulation results are achieved, the designed

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controllers are tested in connection with the real plant by means of RCP. To realize these tests, RCP-technology supplied by MathWorks and dSPACE is utilized. Measurement data can be recorded for post-processing or as stimuli for later offline simulation. By SIL simulation on a host computer, the behavior of the generated code can be compared with the reference data obtained in MIL simulation. The obtained signals should be consistent with the results of MIL simulation.

- Processor in the Loop: Processor in the Loop: in PIL testing the autogenerated code is compiled with the target compiler and downloaded to the evaluation board. Usually the target ECU communicates with the model on the host computer via a serial communication link. PIL testing provides feedback about the required CPU-time for the execution of single control functions and the required resources.
- Hardware in the Loop: in this type of testing the students execute the generated control algorithm on the target ECU that contains sensors and actuators, as well as the simulated plant running on a real-time simulator. The measured data that is recorded in the RCP phase can be used as stimuli. HIL enables the systematic check of all possible events and the response of the ECU. All external IOs, sensors and actuators, are connected to the evaluation board.

C. Realisation Test

After passing all simulated tests the final step is to perform the tests with a real ECU and real plant. These test patterns apply for validation of the ECU, and the measured data can be recorded for post-processing.

II. Students' Projects

For the parameter identification and practical tests several Throttle-Valve Demonstrator Units (TVDU) are built. The TVDUs include an electronic throttle valve (ETV), the driver circuit consisting of a full H-Bridge and the further signal conditioning circuit. The signal conditioning circuit adopts the position sensor voltages to the unified voltage range and enables the measurement of the armature current. Electrical connections of the TVDU include among others the pulse-width-modulated voltage input for the H-Bridge (separated PWM-signal and direction), the analogous outputs of the redundant position sensors and the armature current sensors' output. The analogous electrical signals are all in range of [-10V…10V] whereas the digital signals are TTL-compatible. On this way both RCP-Hardware as well as CPU- and FPGA-based evaluation boards can be used for control.

Fig. 2. Photograph of the throttle valve demonstrator unit.

The next step is the derivation of the linearized model of the throttle valve. The interaction between the electrical and mechanical domain followed by the representation by frequency transfer function is the first goal. Here, the different dynamic properties (e.g. time constants) of the mechanical and electrical subsystem are observed.

A. Parameter Identification

In scope of a student's project [3] the modelling and parameter identification of the electronical throttle valve is performed. Here, the top-down modelling approach is utilized by representation of the plant by an electrical and mechanical subsystem with mutual interactions as shown in Fig. 3.

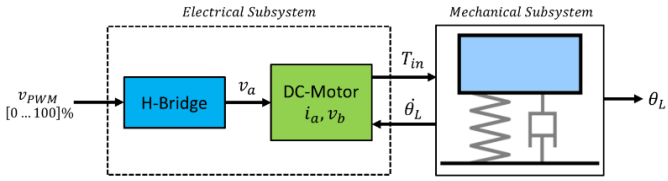

Fig. 3. Representation of plant by electrical and mechanical subsystem.

In Fig. 3, is the v_{PWM} duty-cycle, v_a the resulting state-space averaged voltage, i_a the averaged current, T_{in} the resulting active torque and θ_L the throttle-valve angle position. The physical model of the plant is shown in Fig. 4.

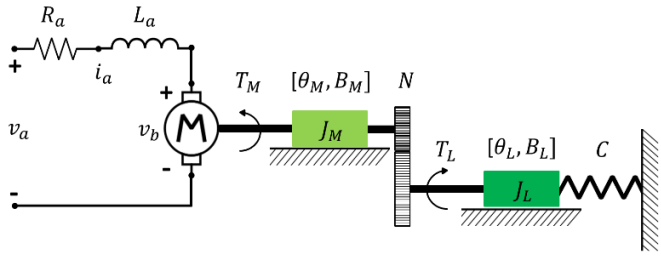

Fig. 4. Representation of plant by electrical and mechanical subsystem.

The mechanical subsystem of the model can be represented as input force, acting upon one single mass with spring and damping characteristics. The static and viscous frictions are represented as the total ones in the system. The armature current in the dc drive produces the acting torque which is here the input variable. When an efficient and fast current-controller is used – implemented in the scope of another student's project [4] – the influence of the mechanical system onto the electrical is eliminated. The resulting simplified mechanical model is shown in Fig. 5.

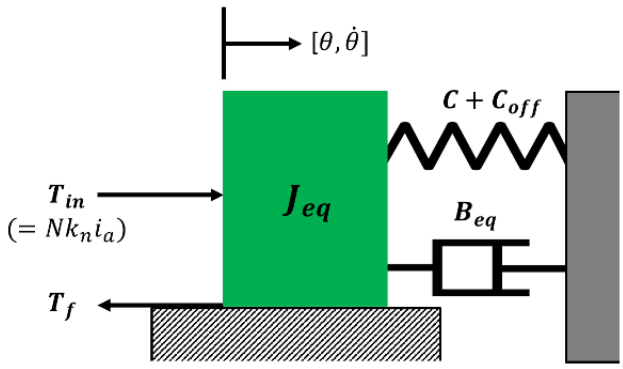

Fig. 5. The simplified model of the mechanical subsystem.

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In Fig. 5. C is the spring torque, C_{off} is the preloaded spring torque, B_{eq} is the equivalent damping, N is the gear ratio, k_n is the motor torque constant and T_f is the static friction. The model is represented as a continuous transfer function.

$$\frac{\theta(s)}{i_a(s)} = \frac{k_n \cdot N}{J_{eq} \cdot s^2 + B_{eq} \cdot s + C} \tag{1}$$

The transfer function parameters are identified using parametric and non-parametric identification methods.

1) *Spring and Preloaded Spring Torque*

For a slowly excited throttle valve, the simplified torque balance equation is represented as:

$$T_{in} = C \cdot \theta + C_{off} + \operatorname{sgn}\left(\frac{d\theta}{dt}\right) \cdot T_f \tag{2}$$

The real throttle position under a ramped input is plotted in Fig. 6. From the plot, the value of C i.e. slope of either line is found to be C = 0.12 Nm/rad. The value of C_{off} = 0.353 Nm. The static friction value is found to be T_f = 0.267 Nm (calculated as 0.62 Nm–0.353 Nm).

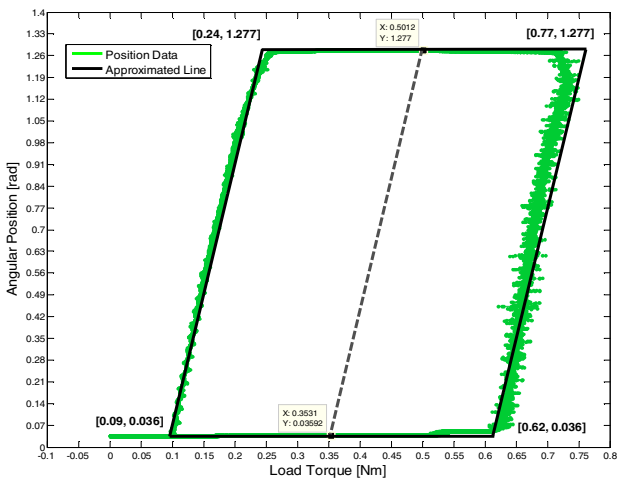

Fig. 6. The measured position of the throttle versus the input torque.

2) *Static Friction and Damping*

Using step input torque excitation, the angular position of the throttle is recorded, and the angular velocity is calculated numerically.

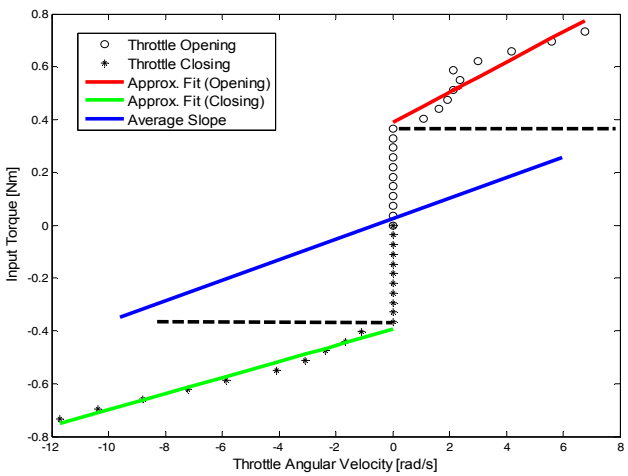

Fig. 7. The angular velocity of the throttle versus the input torque.

At maximum and minimum angular velocity of the throttle, the acceleration is zero, so, the torque balance equation is represented as:

$$T_{in} = B_{eq} \cdot \frac{d\theta}{dt} + C \cdot \theta + C_{off} + \operatorname{sgn}\left(\frac{d\theta}{dt}\right) \cdot T_f \tag{3}$$

The actual torque acting on the throttle is hence:

$$\sum T = B_{eq} \cdot \frac{d\theta}{dt} + \operatorname{sgn}\left(\frac{d\theta}{dt}\right) \cdot T_f \tag{4}$$

Fig. 7. Shows the throttle angular velocity plotted versus the acting torque collected from multiple measurements. The average viscous damping B_{eq} (blue line in Fig. 7) was found to be 0.0042 Nm.

3) *Equivalent Moment of Inertia*

For model identification in frequency domain, it is necessary to deduce the transfer characteristic of the system in a broad frequency spectrum of relevant frequencies. The transfer function of a SISO system can be expressed from the power spectral density of the auto-correlation P_{xx} and cross-correlation P_{xy} of the input and output signal [5]:

$$G_{estimated}(j\omega) = \frac{P_{xy}(j\omega)}{P_{xx}(j\omega)} \tag{5}$$

The ETV is excited by a sine sweep excitation signal with frequency varying from 0.05 rad/s to 300 rad/s and the throttle position response is recorded. The parametric transfer function is fitted with the estimated transfer function from the measured position and the value of moment of inertia of the ETV was found to be: J_{eq}=0.0026 kg□m^2.

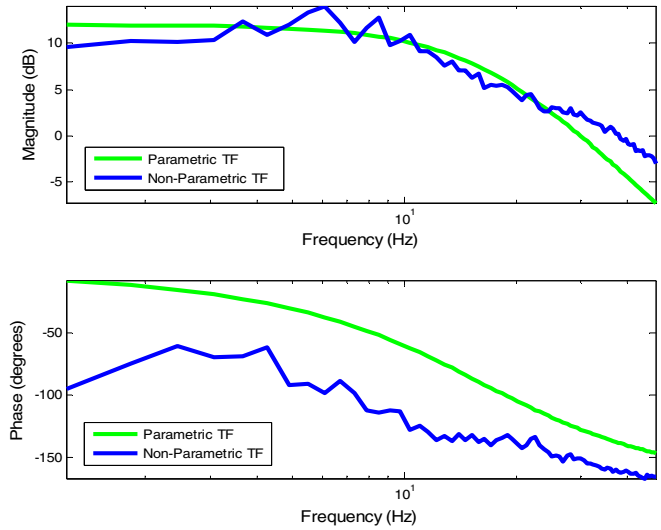

Fig. 8. Frequency domain estimation of moment of inertia in ETV.

B. *Verification and Validation of the ETV-model*

To validate the modelled plant and identified parameters the simulated plant has been compared with the real plant with a step reference input, using a simple PI controller. Using Simulink Coder and dSPACE Real-Time-Interface [6] an executable real-time application is created. The automatically created real-time application is executed on a dSPACE MicroLabBox (MLBX) [7], which is connected to the TVDU.

It can be seen in Fig. 9. that the simulation results obtained in Simulink and the measured real-time control results are very similar.

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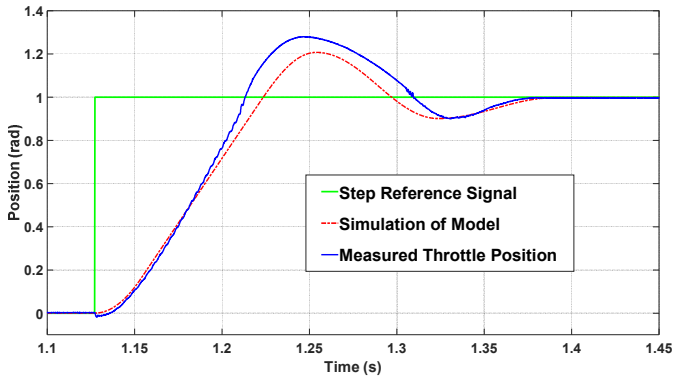

Fig. 9. The measured position of the throttle versus the input torque.

C. Position Control of the Throttle Valve

The design of an appropriate position controller for the ETV was processed in a student project. A controller design by Input-Output Linearization Method was performed. The designed controller was used with the underlying current-controller, which was realized with the FPGA on the MLBX.

In the input-output linearization approach, the objective is to linearize the map between the transformed inputs and the actual output through a change of variables [8]. For this purpose, we use the Lie-Derivative, that as gradient of a scaler function. This part is an implementation of the throttle controller model proposed in [9]. The applied method proposes a control system in form:

$$\begin{aligned} \dot{x} &= a(x) + b(x)\cdot u \\ y &= c(x) \end{aligned} \tag{6}$$

Thus, the system is not linear in x, but linear in u.

The goal is to find a one-to-one state coordinate transformation that transforms the above system in nonlinear control form into a linear one. For this purpose, the Lie-Derivative will be used, which is defined as a gradient of a scalar function [10].

If $x_1(t)$, $x_2(t)$ represent the angular position and the angular velocity of the throttle plate; $u(t)$ is the armature current applied to the DC motor. These equations can be written as:

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= \frac{1}{J}\left(-B\cdot x_2 - C\cdot x_1 - T_{pl} - T_f\cdot \mathrm{sgn}(x_2) + N\cdot k_n\cdot i_a\right) \end{aligned} \tag{7}$$

The model can be expressed in form of nonlinear state space:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} x_2 \\ -\frac{1}{J}\left(B\cdot x_2 + C\cdot x_1 + T_{pl} + T_f\cdot \mathrm{sgn}(x_2)\right) \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{N\cdot k_n}{J} \end{bmatrix}\cdot u \tag{8}$$

$$y = x_1$$

Using input-output feedback control method, we obtain:

$$\begin{aligned} \dot{y} &= L_a\cdot c(x) + L_b\cdot c(x)\cdot u(x) = x_2 \\ \ddot{y} &= \frac{1}{J}\left(-B\cdot x_2 - T_f\cdot \mathrm{sgn}(x_2) - C\cdot x_1 - T_{pl} + N\cdot k_n\cdot u\right) \end{aligned} \tag{9}$$

Now, the closed-loop controller rules can be written in form of $u(x, w) = -r(x) + v(x)\square w$. Since the relative grad $\delta=2$, the transfer function of control loop has as well the order of two, therefore follows:

$$\begin{aligned} G(s) &= \frac{\lambda_1\cdot\lambda_2}{s^2 + (\lambda_1+\lambda_2)\cdot s + \lambda_1\cdot\lambda_2} \\ r(s) &= \frac{\frac{1}{J}\left(-B\cdot x_2 - T_f\cdot \mathrm{sgn}(x_2) - C\cdot x_1 - T_{pl}\right) + a_1\cdot x_2 + a_0\cdot x_1}{N\cdot k_n}\cdot J \\ v(x) &= \lambda_1\cdot\lambda_2\cdot\frac{J}{N\cdot K}. \end{aligned} \tag{10}$$

A good response in simulation can be achieved by placing a pair of complex conjugate poles at $\delta_1 = -60 + 30j$, $\delta_2 = -60 - 30j$. To remove oscilation in steady-state, a smoother approximation for signum function should be used. Therefore, the smooth function $tanh(x\cdot k)$ is used, where parameter k controls the smoothness of the sign function.

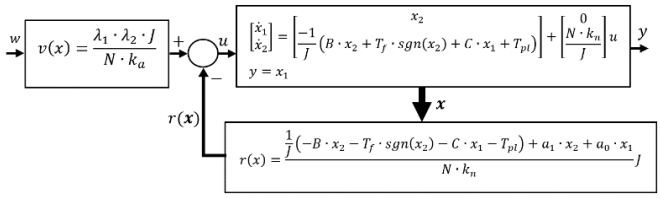

Fig. 10. Action-diagram of the control system with input-output linearization.

D. Model in the Loop

MIL simulation (floating-point) serves as a reference for subsequent steps and provides the minima and maxima of variables as a basis for subsequent fixed-point scaling if required.

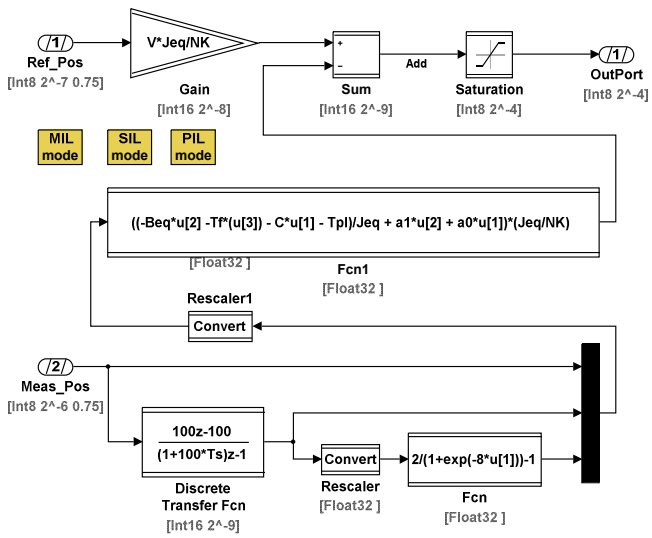

Fig. 11. Controller implementation with TargetLink blocks with scaling.

Automatically generated test case can be applied and stimulate the controller and plant system as input of the position reference. Hear was executed three wave form and it shows that the Simulink and the MIL results were similar, and the performance is satisfying with settling time t_s<150 ms, and no overshoot.

E. Software in the Loop

After satisfactory MIL tests of the designed controllers, the executable object code was generated by TargetLink [11]

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for the subsequent SIL testing. Hire the three recorded results of MIL simulation are used as scaling method. Later, the proper data types have been selected to optimize performance time and accuracy. The raw values of data (not scaled) are considered as inputs of functions in TargetLink, therefore they need rescaler for data type conversion.

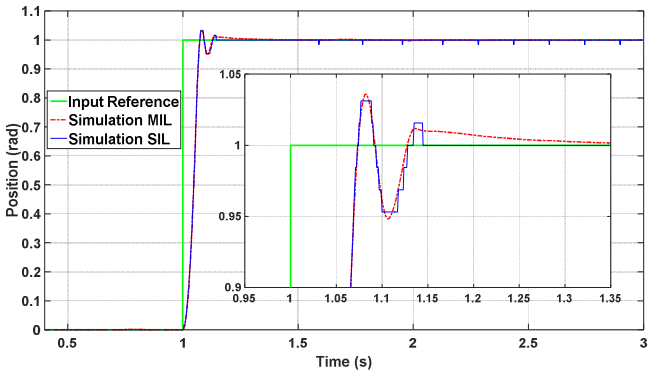

Fig. 12. Comparison between MIL and SIL.

The SIL test performs the computations in fixed-point arithmetic, which reflects the actual computations as they are to be done in the target system. In this test overflows can be detected. Fig. 12. shows the simulation result of MIL for a step input, together with the result simulated using SIL. The MIL and SIL results were practically identical.

F. Processor in the Loop

Finally, the generated code is executed on an evaluation board. Code that performs correctly on a PC may still cause problems with the target processor due to compiler bugs or missing support for the used data types. To validate the correctness of the control running on the target processor, PIL simulation is performed.

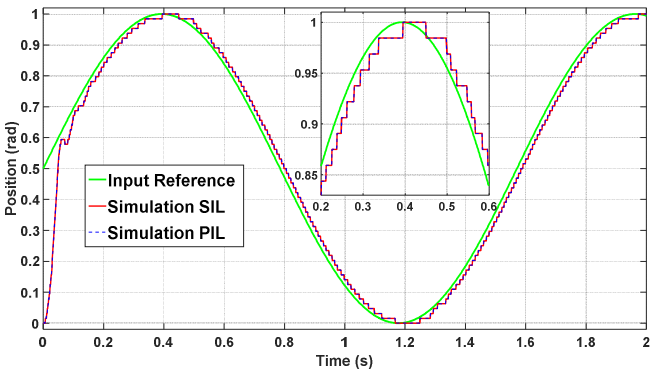

Fig. 13. Comparison between PIL and SIL.

As a target ECU a Renesas RL78-R5F10PP microcontroller that is pre-programed with a TargetLink boot-loader is chosen. IAR Embedded Workbench 1.3 is used as target-compiler. The PIL simulation also provides information on the code size, the required RAM/ROM, and the dynamic stack consumption. The execution time of the function can be displayed as well. Fig. 13. shows that the throttle position of SIL and PIL simulation are identical with sinusoidal input reference.

G. Hardware in the Loop Implementation

With HIL testing the plant (ETV) is replaced by a simulation of the plant running using the MLBX. A well-designed plant simulator used for testing of the controller follows accurately the plant's response. The cascade position controller is executed on a Xilinx FPGA based development board Nexys4. A potentiometer is used as an input reference. The armature current and the throttle position are the outputs of the HIL. These values are provided through the MLBX running the simulated plant with a sample time of 50 µs. The power circuit of DC motor is realised with a full bridge driver. The PWM signal with the duty cycle of 50% equals to an average voltage of zero, the 100% and 0% output correspond to 13.8 V and−13.8 V resp.

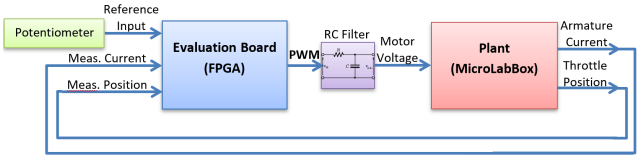

Fig. 14. Set up connection for HIL implementation.

The PWM output of the controller board is applied to the analog input of MLBX. For this purpose, simple RC low-pass-filter is used (with 1 ms time constant).

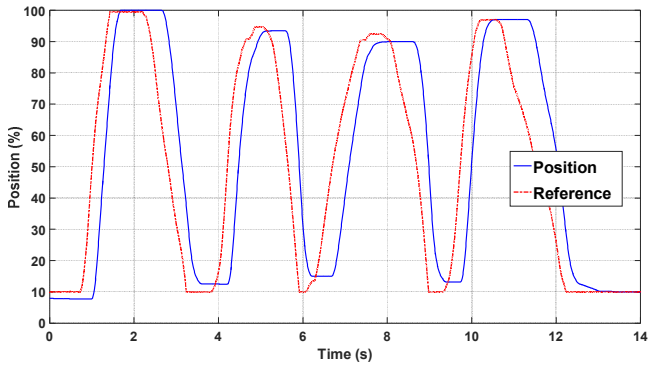

Fig. 15. HIL result for an arbitrary input reference.

The appropriate scaling of signals is considered during HIL testing by considering the signal levels of the evaluation board. The PWM signal generated by the FPGA has the range [0…3.3 V], e.g.−1.65 V offset and 13.8/3.3 V/V gain. The measured sensor voltage of the throttle valve in closed position is 1.66 V, and for completely open 10.07 V, hence the proper equivalent voltage for the position will be calculated as $1.66V+\theta\cdot 5.3$ V/rad. The measured current from the power circuit is converted to voltage with the 1 V/A scaling.

After setting up the combination of HIL, the measured motor voltage and the simulated position and armature current values can be recorded in ControlDesk to validate the realized controller. As input reference, the potentiometer is used which is set manually and the simulated position through the MLBX is observed. From Fig. 15 it can be noticed that the throttle position follows the reference input.

III. Controller Realization and Testing Hardware

A. Hardware Platform

Using the results from the system modelling phase and the realized current controller with the dSPACE MLBX, an FPGA implementation of the ETV position controller with underlying current controller is developed in scope of a student project [12]. As a production hardware platform, the

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Nexys4 development board by Digilent Inc. is used [13], featuring a Xilinx Artix-7 FPGA and several peripheral elements such as switches and seven-segment displays which are incorporated into the design. A further feature of the Nexys4 is an on-chip dual channel 12-bit ADC (XADC), which is utilized for current measurement. To expand the capabilities of the development board external peripherals (Pmods) can be connected, in this use-case an additional ADC in form of a Pmod is added. This Pmod ADC is the AD-2 also supplied by Digilent Inc., featuring four independent 12-bit channels with a typical conversion time of 1 µs. Incorporation of two ADC ensures scalability for future modifications to the design, enhancing the capabilities of the prototype system.

B. Implementation

The implementation of the controller modules and the PWM-generation is performed using XSG in an automated way. The Top Level and I/O processing is realized using VHDL.

C. Realization and Testing

During the implementation of the FGPA design, it became apparent that it is most vital to ensure the correct function of the individual modules by elaborate tests and measurements of signals. To carry out these tests, it was necessary to implement top-level designs for each module itself, to be able to verify the correct operation of the stand-alone modules separately. By conducting extensive module tests and detailed planning for the top-level design the integration phase proved to be manageable without further modifications to the design, making the stand-alone module tests highly worthwhile. To be able to semi-automate the process of implementation of the Simulink model as an FPGA-design the stand-alone function of the peripheral modules must be ensured. After completing the integration via VHDL, an alternative approach could be utilized for future work in form of a block-design, being able to visualize the connections of the modules and reducing the difficulty of troubleshooting. For this approach, IP-Cores would have to be generated for the stand-alone modules using Vivado and Simulink for the controller.

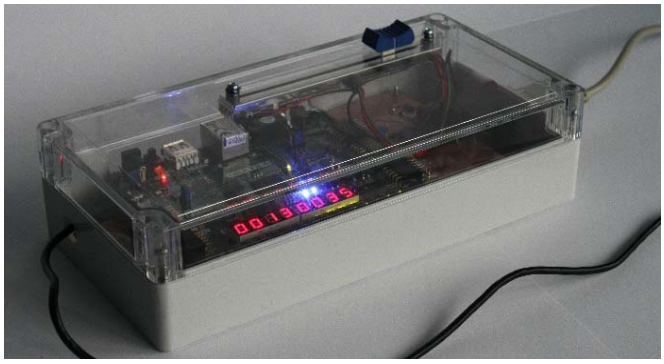

Fig. 16. ETV-Controller realization with Nexys4 FPGA evaluation-board.

The generated cores could then be integrated in a block-design, in which the controller IP-Core can be substituted in less time, being able to concentrate more on the controller design than the function of the peripherals. The ETV-controller realized with the Nexys4 FPGA EVB is shown in Fig. 16. The FPGA controller board was evaluated using an arbitrary input signal (according to the signals obtained in the HIL test). The system response is shown in Fig. 17. The opening of the ETV is slightly slower in practice, but faster in closing. The test shows that the controller achieves satisfying results when applied to the physical plant.

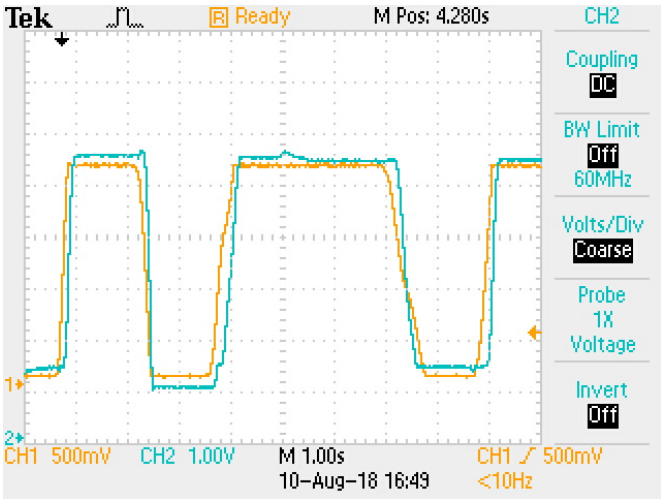

Fig. 17. Experimental measurement. The differing voltage settings are due to different voltage sources for the individual components.

IV. Conclusion

MDE-based methods are a commonly used practice in the industry, and these methods are now a part of the study plan of THD. Some courses are designed to prepare engineers for industry requirements. In this paper we gave a short overview about related activities on THD. In our future work, we will broaden the field of applications and include hybrid CPU/FPGA platform in MDE education as well.

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